gpeyt1-17 · FS against liquefaction

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

A silty sand layer (FC = 12%) at 18 ft has a clean-sand-equivalent (N1)60cs = 13, giving CRR(M7.5) = 0.17 from the base curve. The design event is Mw = 6.75 with amax = 0.28g; the magnitude scaling factor for that event is MSF = 1.19. Kσ = 1.0. Water table 8 ft; γ = 115 pcf above, γsat = 122 pcf below. Use rd = 1 − 0.00233z.

Most nearly, the factor of safety against liquefaction is:

A. 0.69
B. 1.22
C. 1.16
D. 0.82

Given equations / data

> CSR = 0.65 (amax/g)(σvo/σ′vo) rd; FSL = (CRR7.5 · MSF · Kσ)/CSR

Worked solution

1. σvo = 8(115) + 10(122) = 2,140 psf; σ′vo = 1,516 psf
2. rd = 0.9581; CSR = 0.65×0.28×1.412×0.9581 = 0.246
3. FSL = 0.17×1.19×1.0/0.246 = 0.82 (<1 → liquefiable)

Final answer

0.82

Why the other values are wrong

- **0.69** — omitted MSF
- **1.22** — inverted FS (CSR/CRR)
- **1.16** — used total stress for σ′vo (ratio = 1)

References (Jayce's library)

- FHWA-NHI-11-032 (GEC 3) Ch. 6 Eq. 6-16 and FS definition, MSF discussion

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=Hg3MTAvoJWw (no creator values copied).

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